

Quiz details

Quiz instructions:

This quiz will focus on the material from Week 10 in the lab workbook. To answer the following questions, you will need to use the [Sedum_traits.csv](https://canvas.stir.ac.uk/courses/19303/files/4997300/download?download_frd=1) [↓](https://canvas.stir.ac.uk/courses/19303/files/4997300/download?download_frd=1) (https://canvas.stir.ac.uk/courses/19303/files/4997300/download?download_frd=1) dataset.

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Sedum dataset

The genus *Sedum* (family Crassulaceae) includes over 600 species of flowering plants. You have collected trait data from 3 species of *Sedum* in the UK, including *S. acre*, *S. album*, and *S. sexangulare*. You want to better study how the traits of 3 species vary. Here are the columns of data in your dataset

([Sedum_traits.csv](https://canvas.stir.ac.uk/courses/19303/files/4997300?wrap=1) (<https://canvas.stir.ac.uk/courses/19303/files/4997300?wrap=1>) [↓](https://canvas.stir.ac.uk/courses/19303/files/4997300/download?download_frd=1) (https://canvas.stir.ac.uk/courses/19303/files/4997300/download?download_frd=1)) :

1. Plant
2. Species
3. Habitat
4. Leaf_Area (mm²)
5. Leaf_Thickness (cm)
6. Seed_dry_mass (μg)
7. Root_radius (mm)
8. Leaf_Carbon (%)
9. Leaf_Nitrogen (%)
10. Leaf_Phosphorus (%)
11. Plant_height (cm)
12. Leaf_dry_matter (leaf dry mass per unit leaf fresh mass)
13. Altitude (m)
14. Latitude

Use the data in these datasets to answer the following questions.

Question 1 1 pts

Assume that Plant_height is normally distributed. Run a linear regression to predict change in Plant_height given a change in Altitude. What percentage of the variation in the dependent variable is explained by the model? Report your answer to 2 or more decimal places.

9.65 (with margin: 0.02)

Correct answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Plant_height' in the Dependent Variable box, then put 'Altitude' in the Covariates box. Under the 'Model Fit' pulldown below, make sure that R² is selected. To the right, there will be a table called 'Model Fit Measures'.

Model Fit Measures		
Model	R	R ²
1	0.31065	0.09650

The value 0.09650 from this table is the proportion of the variation explained by the model. To get this as a percentage, you needed to multiply by 100, **0.09650 × 100 = 9.650**

Wrong answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Plant_height' in the Dependent Variable box, then put 'Altitude' in the Covariates box. Under the 'Model Fit' pulldown below, make sure that R² is selected. To the right, there will be a table called 'Model Fit Measures'.

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The value 0.09650 from this table is the proportion of the variation explained by the model. To get this as a percentage, you needed to multiply by 100, **0.09650 × 100 = 9.650**

Question 2 1 pts

Again use the same linear model from Question 1 predicting change in Plant_Height given a change in Altitude. What is the slope of the regression line of this model? Report your answer to 3 or more decimal places.

-0.0417 (with margin: 0.002)

Correct answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Plant_height' in the Dependent Variable box, then put 'Altitude' in the Covariates box. In the panel to the right a table called 'Model coefficients' will appear. The slope can be found under the 'Estimate' column.

Model coefficients table from Jamovi

Model Coefficients - Plant_height

Predictor	Estimate	SE	t	p
Intercept	18.84517	2.26807	8.30891	<.00001
Altitude	-0.04168	0.00906	-4.59870	<.00001

The slope is in the second row called 'Altitude', so the slope estimate is -0.04168.

Wrong answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Plant_height' in the Dependent Variable box, then put 'Altitude' in the Covariates box. In the panel to the right a table called 'Model coefficients' will appear. The slope can be found under the 'Estimate' column.

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Altitude	-0.04168	0.00906	-4.59870	<.00001

The slope is in the second row called 'Altitude', so the slope estimate is -0.04168.

Question 3 1 pts

Now, run a linear regression with Latitude as your independent variable and Root_radius as your dependent variable. Which of the following can you conclude about your model coefficients based on your Jamovi output?

- ☐ Both the intercept and the slope are significantly different from zero
- ☐ The intercept is significantly different from zero, but the slope is not significantly different from zero
- ☐ The intercept is not significantly different from zero, but the slope is significantly different from zero
- ☐ Neither the slope nor the intercept are significantly different from zero

Correct answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Root_radius' in the Dependent Variable box, then put 'Latitude' in the Covariates box. A table called 'Model Coefficients - Root_radius' will appear in the panel to the right.

Jamovi model coefficients table

Model Coefficients - Root_radius

Predictor	Estimate	SE	t	p
Intercept	0.49498	0.03333	14.85161	<.00001

Model Coefficients - Root_radius

Predictor	Estimate	SE	t	p
Latitude	0.00079	0.00062	1.28691	0.19963

From the p-values in the last column (p), we can see that we should reject the null hypothesis that the intercept equals zero because $P < 0.05$. But we should not reject the null hypothesis that the slope equals zero because $P > 0.05$. We should therefore conclude that the intercept is significantly different from zero, but the slope is not significantly different from zero.

Wrong answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Root_radius' in the Dependent Variable box, then put 'Latitude' in the Covariates box. A table called 'Model Coefficients - Root_radius' will appear in the panel to the right.

Jamovi model coefficients table

Model Coefficients - Root_radius

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Question 4 1 pts

Use the same linear regression as in Questions 1 and 2, with Altitude as your independent variable and Plant_height as your dependent variable. Based on the regression coefficients you obtain in Jamovi, predict Plant_height given an Altitude of 260 metres. Report your answer to 2 or more decimal places.

8.0084 (with margin: 0.02)

Correct answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Plant_height' in the Dependent Variable box, then put 'Altitude' in the Covariates box. In the panel on the right, a Model Coefficients - Plant_height table will appear.

Jamovi model coefficients table

Model Coefficients - Plant_height

Predictor	Estimate	SE	t	p
Intercept	18.84517	2.26807	8.30891	<.00001
Altitude	-0.04168	0.00906	-4.59870	<.00001

The table gives estimates for the model intercept ($b_0 = 18.84517$) and slope ($b_1 = -0.04168$). We can use these in the equation from the work book [Chapter 31.3](https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-coefficients) [↗](https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-coefficients) (https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-coefficients),

$$y = b_0 + b_1x.$$

We have b_0 and b_1 , and the question tells the value of $x = 260$. We can therefore substitute this information,

$$18.84517 = (0.04168 \times 260).$$

This gives us the final value of 8.00837.

Wrong answer comments

To run a linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable

'Plant_height' in the Dependent Variable box, then put 'Altitude' in the Covariates box. In the panel on the right, a Model Coefficients - Plant_height table will appear.

Jamovi model coefficients table

Model Coefficients - Plant_height

Predictor	Estimate	SE	t	p
Intercept	18.84517	2.26807	8.30891	<.00001
Altitude	-0.04168	0.00906	-4.59870	<.00001

The table gives estimates for the model intercept ($b_0 = 18.84517$) and slope ($b_1 = -0.04168$). We can use these in the equation from the work book [Chapter 31.3](https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-coefficients) (https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-coefficients),

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We have b_0 and b_1 , and the question tells the value of $x = 260$. We can therefore substitute this information,

$$18.84517 = (0.04168 \times 260).$$

This gives us the final value of 8.00837.

Question 5 1 pts

Again, use the same linear regression as in Questions 1 and 2, with Altitude as your independent variable and Plant_height as your dependent variable. Suppose you were to go for a hike that increased your altitude by 10 metres. How much of a change in Plant_height would you predict in the *Sedum* species you see at the end of your hike?

Report your answer to 2 or more decimal places.

-0.4168 (with margin: 0.02)

Correct answer comments

From the previous question we found that the slope of Altitude was -0.04168. This tells us how much change there will be in our dependent variable (Plant height) given a unit change in the independent variable (Altitude). In other words, if we increase Altitude by 1, we expect a change in Plant_height of -0.04168. To find how much change we would expect given 10 metres increase in Altitude, we just need to multiply -0.04168 by 10, $-0.04168 \times 10 = -0.4168$.

Wrong answer comments

From the previous question we found that the slope of Altitude was -0.04168. This tells us how much change there will be in our dependent variable (Plant height) given a unit change in the independent variable (Altitude). In other words, if we increase Altitude by 1, we expect a change in Plant_height of -0.04168. To find how much change we would expect given 10 metres increase in Altitude, we just need to multiply -0.04168 by 10, $-0.04168 \times 10 = -0.4168$.

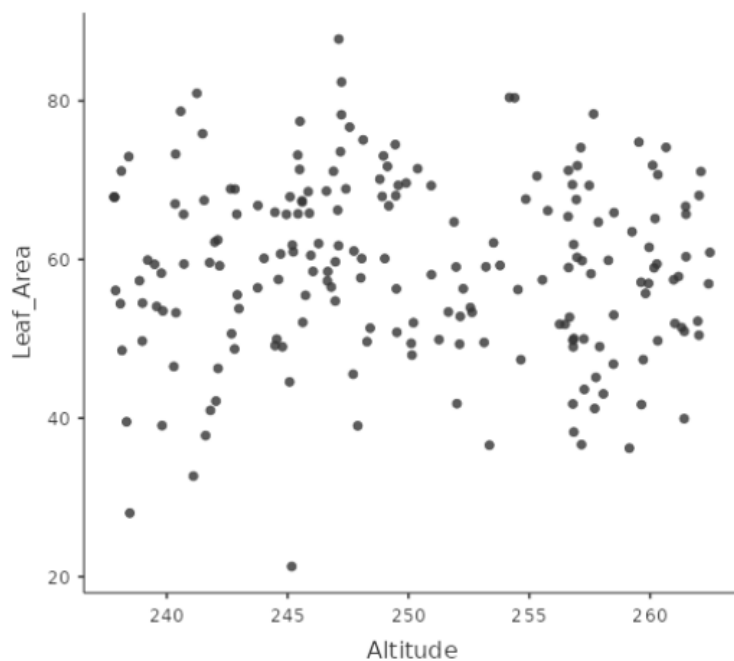
Question 6 1 pts

Plot a fitted regression line with Altitude as your independent variable and Leaf_Area as your dependent variable. For which altitude would you be most uncertain about predicting Leaf_Area and why?

- ☐ 240 m because this would be extrapolating outside the measurement range of the data
- ☐ 250 m because this would be extrapolating outside the measurement range of the data
- ☐ 260 m because this would be extrapolating outside the measurement range of the data
- ☐ 280 m because this would be extrapolating outside the measurement range of the data
- ☐ 240 m because this would be interpolating too much within the measurement range of the data
- ☐ 250 m because this would be interpolating too much within the measurement range of the data
- ☐ 260 m because this would be interpolating too much within the measurement range of the data
- ☐ 280 m because this would be interpolating too much within the measurement range of the data

Correct answer comments

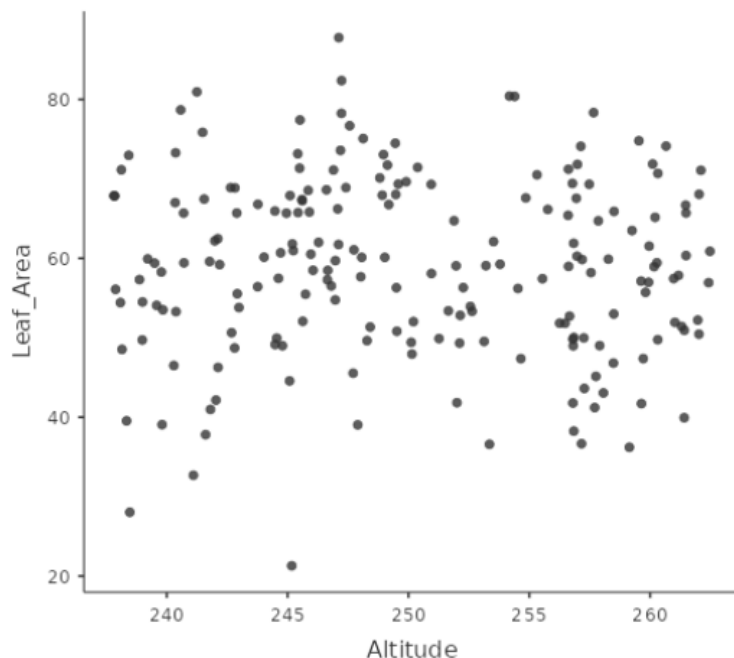
If you use a scatterplot to plot the independent variable 'Altitude' on the x-axis and the dependent variable 'Leaf_Area' on the y-axis, then you will see a figure like the one below.



Notice that the values of altitude range from about 230 to about 270. If we were to make a prediction for an Altitude outside of this range, then we would be extrapolating outside of the measurement range of the data. This would be a problem (see the workbook [Chapter 31.8](#) https://bradduthie.github.io/SCIU4T4/Chapter_31.html#prediction-with-linear-models).

Wrong answer comments

If you use a scatterplot to plot the independent variable 'Altitude' on the x-axis and the dependent variable 'Leaf_Area' on the y-axis, then you will see a figure like the one below.



Notice that the values of altitude range from about 230 to about 270. If we were to make a prediction for an Altitude outside of this range, then we would be extrapolating outside of the measurement range of the data. This would be a problem (see the workbook [Chapter 31.8](#) https://bradduthie.github.io/SCIU4T4/Chapter_31.html#prediction-with-linear-models).

Question 7 1 pts

Now suppose that you want to run a simple linear regression to predict Leaf_dry_matter from Latitude. Which of the following assumptions would you be violating? Check all that apply.

☐ The relationship between the independent and dependent variables is linear

- ☐ For all values of the independent variable, the variance of the residuals is identical
- ☐ The adjusted coefficient of determination (adjusted R-squared) does not exceed the unadjusted coefficient of determination
- ☐ The independent variable is normally distributed around the regression line
- ☐ None, because regression assumptions only apply to multiple linear regression, not simple linear regression

Correct answer comments

There are 3 potential assumptions to check (see [Chapter 31.6](https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-assumptions) https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-assumptions), note that we cannot test the first assumption in this section, so we have to assume that independent variables were measured accurately). First, we can check to see that residuals are normally distributed around the regression line. To do this, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Leaf_dry_matter' in the Dependent Variable box, then put 'Latitude' in the Covariates box. Make sure that 'Normality test' is selected from the 'Assumption Checks' pulldown menu. From the panel on the right, we can see that based on the Normality Test, we get a p-value of 0.00125.

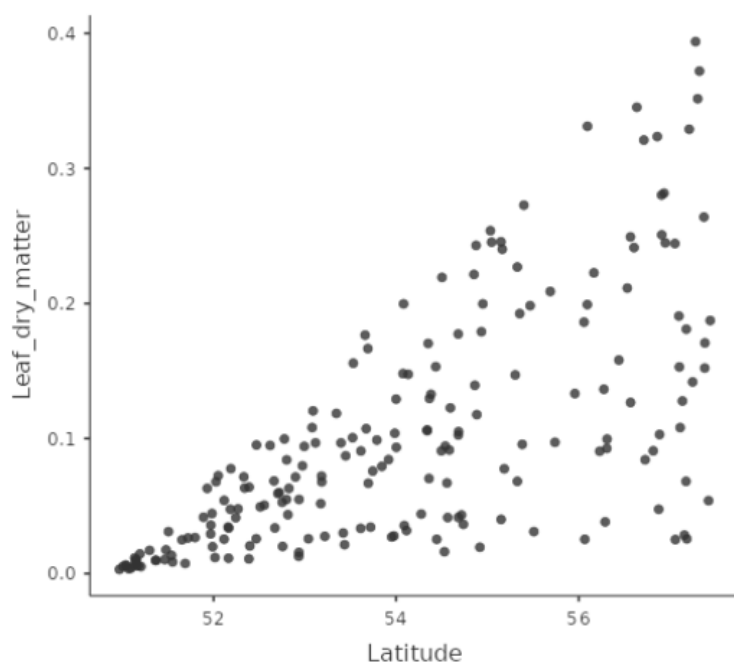
Jamovi normality test

Normality Test (Shapiro-Wilk)

Statistic	p
0.97506	0.00125

Consequently, if we ran a linear regression, we would be violating the assumption of normality.

Next, we should use a scatterplot to check that the relationship between independent and dependent variables is linear and the variance of the residuals is identical. Navigate to 'Exploration' and Scatterplot to build the scatterplot.



While the relationship appears to be linear, the variance in leaf dry matter clearly increases with increasing latitude. We would be violating the assumption of homoscedasticity (identical variance) if we ran a linear regression.

Wrong answer comments

There are 3 potential assumptions to check (see [Chapter 31.6](https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-assumptions) https://bradduthie.github.io/SCIU4T4/Chapter_31.html#regression-assumptions), note that we cannot test the first assumption in this section, so we have to assume that independent variables were measured accurately). First, we can check to see that residuals are normally distributed around the regression line. To do this, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Leaf_dry_matter' in the Dependent Variable box, then put 'Latitude' in the Covariates box. Make sure that 'Normality test' is selected from the 'Assumption Checks' pulldown menu. From the panel on the right, we can see that based on the Normality Test, we get a p-value of 0.00125.

Jamovi normality test

Normality Test (Shapiro-Wilk)

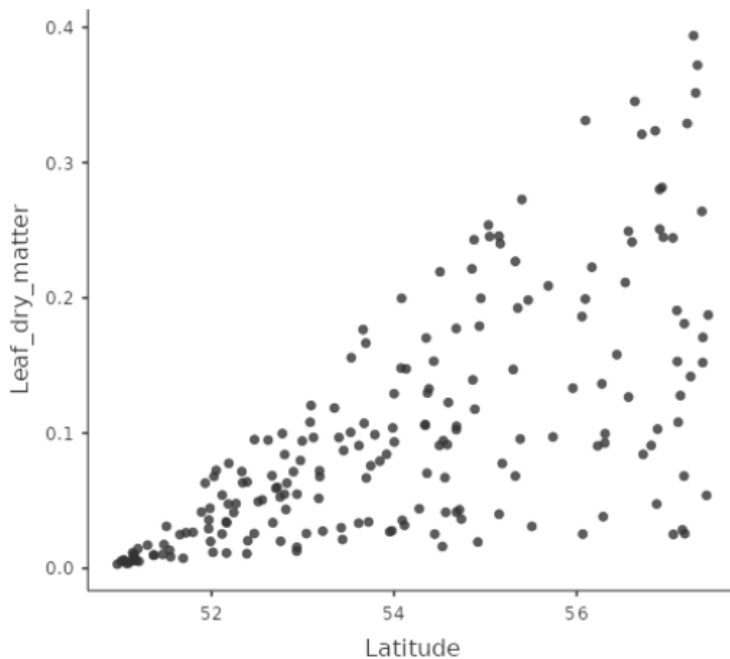
Statistic	p
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Normality Test (Shapiro-Wilk)

Statistic	p
0.97506	0.00125

Consequently, if we ran a linear regression, we would be violating the assumption of normality.

Next, we should use a scatterplot to check that the relationship between independent and dependent variables is linear and the variance of the residuals is identical. Navigate to 'Exploration' and Scatterplot to build the scatterplot.



While the relationship appears to be linear, the variance in leaf dry matter clearly increases with increasing latitude. We would be violating the assumption of homoscedasticity (identical variance) if we ran a linear regression.

Question 8 1 pts

Run a multiple linear regression with `Root_radius` as your dependent variable and `Leaf_Carbon`, `Leaf_Nitrogen`, and `Leaf_Phosphorus` as independent variables. Report the percentage (not the proportion) of variation in `Root_radius` explained by the model, as properly adjusted to account for the spurious improvement of fit caused by multiple independent variables. Report your answer to at least 1 decimal place.

65.37 (with margin: 0.1)

Correct answer comments

To run a multiple linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable '`Root_radius`' in the Dependent Variable box, then put '`Leaf_Carbon`', '`Leaf_Nitrogen`', and '`Leaf_Phosphorus`' in the Covariates box. Find the 'Model Fit' pulldown below, then make sure 'Adjusted R^2 ' is checked. You will see a table called Model Fit Measures below.

Jamovi model fit output

Model Fit Measures

Model	R	R^2	Adjusted R^2
1	0.81174	0.65893	0.65370

The Adjusted R^2 gives us the R^2 adjusted for multiple independent variables. This is the value we need, but we need to multiply it by 100 to get the percentage of the variation explained by the model, $0.65370 \times 100 = 65.37$.

Wrong answer comments

To run a multiple linear regression in Jamovi, navigate to the 'Analyses' tab in Jamovi, then choose the 'Regression' button. Put the variable 'Root_radius' in the Dependent Variable box, then put 'Leaf_Carbon, Leaf_Nitrogen, and Leaf_Phosphorus' in the Covariates box. Find the 'Model Fit' pulldown below, then make sure 'Adjusted R²' is checked. You will see a table called Model Fit Measures below.

Jamovi model fit output

Model Fit Measures

Model	R	R ²	Adjusted R ²
1	0.81174	0.65893	0.65370

The Adjusted R² gives us the R² adjusted for multiple independent variables. This is the value we need, but we need to multiply it by 100 to get the percentage of the variation explained by the model, $0.65370 \times 100 = 65.37$.

Question 9 1 pts

Again use your multiple linear regression from Question 8 with Root_radius as your dependent variable and Leaf_Carbon, Leaf_Nitrogen, and Leaf_Phosphorus as independent variables. Which of the following coefficients of the model are significant? Check all that apply.

- ☐ The intercept
- ☐ Leaf_Carbon
- ☐ Leaf_Nitrogen
- ☐ Leaf_Phosphorus
- ☐ None of the above coefficients are significant

Correct answer comments

From the Model Coefficients - Root radius table, we can see that the p-value for every term in the multiple linear regression is significant because $P < 0.05$. You therefore needed to select everything but the last box.

Wrong answer comments

From the Model Coefficients - Root radius table, we can see that the p-value for every term in the multiple linear regression is significant because $P < 0.05$. We can therefore reject the null hypothesis that the Intercept and all coefficients equal 0. You therefore needed to select everything but the last box.

Question 10 1 pts

Again use your multiple linear regression from Question 8 with Root_radius as your dependent variable and Leaf_Carbon, Leaf_Nitrogen, and Leaf_Phosphorus as independent variables. Considering the coefficient values of your model, predict the Root_radius of a *Sedum* plant with a Leaf_Carbon = 44%, Leaf_Nitrogen = 2%, and Leaf_Phosphorus = 1%. Report your answer to 3 or more decimal places.

0.528 (with margin: 0.001)

Correct answer comments

We need to use the coefficients from the Model coefficients table of the multiple regression model:

We can put these values into an equation,

$$\text{Root radius} = 0.29303 + (0.00608 \times \text{Leaf Carbon}) + (-0.00723 \times \text{Leaf Nitrogen}) + (-0.01811 \times \text{Leaf Phosphorus}).$$

We then just need to substitute the different values from the question,

$$\text{Root radius} = 0.29303 + (0.00608 \times 44) + (-0.00723 \times 2) + (-0.01811 \times 1).$$

Make sure to do the order of operations correctly,

$$\text{Root radius} = 0.29303 + (0.26752) + (-0.01446) + (-0.01811).$$

We can then sum to get our answer,

$$\text{Root radius} = 0.52798.$$

Wrong answer comments

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